

KŁODZKO, Poland

WMO: 125200

Lat: 50.4369N

Lon: 16.6142E

Elev: 1171

StdP: 14.08

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.5	(c) 9.9	(d) 0.6	(e) 5.9	(f) 5.4	(g) 5.6	(h) 7.6	(i) 11.0	(j) 30.8	(k) 32.8	(l) 27.4	(m) 33.7	(n) 2.4	(o) 20	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	84.1	68.0	80.5	66.3	77.4	64.8	69.3	80.8	67.5	78.0	65.9	75.1	7.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	98.2	74.2	63.8	92.5	72.0	62.3	87.6	70.3	34.1	81.0	32.6	78.2	31.3	75.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.7	(b) 20.3	(c) 17.4	(e) -1.6	(f) 89.4	(g) 8.1	(h) 2.8	(i) -7.4	(j) 91.5	(k) -12.2	(l) 93.1	(m) -16.7	(n) 94.7	(o) -22.6	(p) 96.7		
(4) 23.7	(5) 20.3	(6) 17.4	(7) -2.1	(8) 73.1	(9) 8.0	(10) 1.9	(11) -7.8	(12) 74.4	(13) -12.6	(14) 75.5	(15) -17.1	(16) 76.6	(17) -22.9	(18) 78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	28.4	31.1	37.4	46.5	54.2	60.1	63.8	63.4	55.4	47.6	39.3	31.3	(6)
(7)		DBStd	14.41	9.72	9.36	7.76	7.35	6.31	5.90	5.26	5.51	5.93	7.23	7.61	8.79	(7)
(8)		HDD50	2837	670	530	395	150	30	2	0	0	17	134	328	581	(8)
(9)		HDD65	6839	1135	949	857	555	338	169	87	97	293	541	772	1046	(9)
(10)		CDD50	1596	0	0	3	45	160	304	428	415	177	58	6	0	(10)
(11)		CDD65	125	0	0	0	0	3	21	50	47	4	0	0	0	(11)
(12)		CDH74	1531	0	0	0	7	54	259	610	542	59	0	0	0	(12)
(13)		CDH80	353	0	0	0	0	4	45	157	139	8	0	0	0	(13)
(14)	Wind	WSAvg	6.3	7.9	7.7	7.3	6.2	5.4	4.5	4.6	4.5	4.9	6.8	7.7	8.1	(14)
(15)	Precipitation	PrecAvg	32.0	2.2	1.9	2.4	1.9	3.1	3.5	4.4	3.3	2.9	2.1	2.2	2.2	(15)
(16)		PrecMax	39.6	4.3	4.1	5.7	3.4	6.1	6.6	13.4	10.4	5.6	5.3	3.9	4.1	(16)
(17)		PrecMin	22.3	0.6	0.3	0.7	0.4	0.7	1.1	1.0	1.3	0.8	0.3	0.0	0.8	(17)
(18)		PrecStd	4.7	1.0	1.0	1.2	0.8	1.4	1.5	2.6	2.0	1.5	1.2	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	52.9	62.5	74.6	79.0	85.0	89.0	89.2	80.4	70.1	58.6	51.1	(19)
MCWB			46.2	45.8	50.8	58.8	63.8	70.5	68.4	68.8	64.4	59.6	52.8	47.2	(20)	
2%		DB	46.5	49.2	56.9	69.1	74.9	80.5	84.3	83.8	74.5	65.7	55.5	47.9	(21)	
		MCWB	43.4	44.7	47.8	55.6	62.4	68.0	67.7	67.5	61.9	57.4	51.2	44.7	(22)	
5%		DB	43.4	46.2	52.7	64.3	71.2	76.8	80.5	79.8	70.8	62.0	53.0	44.8	(23)	
		MCWB	40.5	42.3	45.9	52.8	60.1	65.3	66.2	65.9	60.2	55.7	49.0	42.1	(24)	
10%		DB	40.4	43.2	49.0	60.4	67.6	73.2	77.0	76.2	67.0	58.9	50.4	42.0	(25)	
		MCWB	38.0	39.9	43.8	50.7	57.8	62.9	64.9	64.1	58.8	53.9	46.9	39.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	47.3	51.6	59.7	66.3	72.5	71.7	70.8	66.3	60.9	53.7	47.5	(27)
MCD B			49.9	51.3	59.7	72.2	76.0	82.3	83.5	83.8	75.8	67.8	57.6	50.5	(28)	
2%		WB	43.4	44.9	48.8	56.6	63.2	69.0	69.1	68.8	63.5	58.4	51.6	44.9	(29)	
		MCD B	46.3	48.8	55.5	67.1	72.5	78.7	80.5	80.8	72.3	64.0	55.3	47.7	(30)	
5%		WB	40.7	42.6	46.5	53.9	61.0	66.2	67.5	67.0	61.4	56.3	49.4	42.1	(31)	
		MCD B	43.3	45.9	51.9	62.9	69.5	74.8	78.0	77.3	68.7	61.1	52.5	44.6	(32)	
10%		WB	38.1	39.9	44.1	51.4	58.8	63.9	65.7	65.2	59.5	54.3	47.0	40.0	(33)	
		MCD B	40.2	42.9	48.6	59.0	65.9	71.5	74.8	74.2	65.9	58.4	50.0	42.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	10.5	13.3	18.1	18.7	18.7	19.6	20.6	16.8	13.3	9.1	8.1	(35)
MCD BR			10.0	14.0	21.6	25.7	25.7	26.1	27.7	27.4	24.4	19.1	12.6	9.4	(36)	
MCWBR			8.1	10.4	14.0	14.5	14.7	14.4	12.9	12.9	13.0	12.0	9.2	7.7	(37)	
5% WB		MCD BR	9.6	12.8	19.4	23.8	23.3	23.9	24.1	24.2	20.6	17.5	11.6	9.2	(38)	
		MCWBR	8.0	10.1	13.1	13.8	14.1	13.9	12.2	12.2	11.9	11.4	8.9	7.8	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.331	0.374	0.423	0.421	0.418	0.437	0.429	0.392	0.365	0.338	0.302	(40)	
taud		2.296	2.267	2.218	2.154	2.208	2.247	2.203	2.238	2.311	2.347	2.361	2.350	(41)		
Ebn at Noon		227	251	259	259	266	267	259	254	252	236	213	211	(42)		
Edh at Noon		22	29	36	43	43	42	43	40	33	27	21	18	(43)		
(44)	All-Sky Solar Radiation	RadAvg	263	492	823	1300	1583	1668	1585	1425	1007	590	290	197	(44)	
(45)		RadStd	48	78	132	169	188	160	185	130	149	87	39	23	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.40	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (46 neighbors)	+0.86	N/A	N/A	N/A	+0.59	+0.54	N/A	N/A	+112	+43	(47)

Nomenclature: See separate page